



A lightweight CNN-based ensemble approach for early detecting Parkinson's disease with enhanced features

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Abstract

Parkinson's disease (PD) presents a neurological challenge affecting individuals with and without motor skills, particularly manifesting as speech difficulties during its initial phases. The successful diagnosis of PD often relies on the detection of these speech problems. Numerous initiatives are still being implemented to solve these problems and improve the accuracy of the diagnosis of PD. However, several computational techniques have been effective in identifying PD in its early stages, bypassing the drawbacks of conventional diagnostic techniques. This study proposed an ensemble approach that combines machine learning (ML) and convolutional neural network (CNN) techniques to detect Parkinson's disease (PD) in its early stages. For our investigation, we used the well-known PD dataset from the University of California, Irvine (UCI) repository. The proposed ensemble approach is developed using two base methods: the multilayer perceptron (MLP) and a hybrid model called ParkRLP. To create the new dataset, the ParkRLP and MLP networks are level-0 learners, while the lightweight PD-CNN model is a level-1 learner. Successful PD detection is achieved by the proposed ensemble strategy using the level-1 classifier as the PD-CNN model. The lightweight PD-CNN model consists of eight layers. The conv-1D, Maxpooling-1D, and Dense layers have been used to modify the PD-CNN model. According to the dependent and independent factors in the newly generated dataset, the meta-classifiers were utilized for training. Then, the classifiers' outputs are determined after they have been trained and tested at level-0. The efficacy of the proposed model was further underscored by its consistent 5% improvement over previous research on individual ML and CNN models. In conclusion, out of all the techniques for detecting PD, the proposed ensemble method of the lightweight CNN approach had the highest accuracy, at 99.47%, and an F1-score of 99.50%. Experimental findings demonstrate the accuracy of the proposed methodology in estimating PD and its simplicity for implementation in medical settings for diagnostic purposes.

Keywords Parkinson disease · CNN · Neural network · Ensemble · SMOTE

1 Introduction

PD is a neurological disease of the autonomic nervous system that results in total or partial damage to voice, behavior, psychomotor skills, and other necessary skills (Costa et al., 2023). Most of the time, PD affects how well a person can move. This can cause someone to start moving slowly. A person with PD appears stiff or restrictive (Jellinger, 2022).

The number of affected people will increase as the world's population ages and deteriorates over time. Although there is no definitive treatment for PD, it can be managed (Costa et al., 2023; Yang et al., 2020).

Approximately a million people in the US have PD. Surgery and prescription drugs can be used to treat these symptoms. There is still no effective treatment for PD (Laux, 2022). This study focused on the diagnostic pathways based on voice analysis. The focus of current PD telemedicine studies is on voice-based systems. Thus, the information required for the assessment of PD is extracted using a variety of speech signal processing algorithms. After early diagnosis, doctors can use medication or central nervous system computation (Mohammadi et al., 2021) to reactivate the brain's dopamine-producing neurons, thereby slowing the progression of PD. Owing to its complexity, there

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is currently no known cure for PD. However, by lessening spasticity and imbalance, early detection and treatment can assist patients in leading normal lives. To create trustworthy decision support systems, the extracted characteristics are fed into learning algorithms. Early diagnosis of PD is now more frequently accomplished using ML and deep learning (DL) techniques. Consequently, there has been a notable improvement in the accuracy of PD forecasting using a range of data methods, including cursive writing patterns (Salari et al., 2023) and, speech-processing signals (Khaskhoussy & Ayed, 2022). Several studies have been conducted on speech to track disease development. ML methods are being continuously applied in the healthcare industry. ML algorithms are used in conjunction with various data modalities, including cursive patterns and acoustic voice recordings, to diagnose PD. With the help of ML techniques, we can find relevant characteristics that are not usually used in the medical diagnosis of PD, and we can rely on these different scenarios to identify PD (Rana et al., 2022).

To increase the accuracy of PD diagnosis, Riyadh et al. (Hashim et al., 2024) utilized well-known machine learning techniques. The authors have proposed several distinctive and ensemble AI models. The stacking Ensemble algorithm which uses ML models to extract features and categorize PD, yields the best results. The accuracy of the Ensemble algorithm achieved 94.87% for the first dataset and 96.18% for the second. Syed et al. (Bukhari & Ogudo, 2024) developed an ML detection technique that uses PD patients' speech signals. The UCI dataset was used to train the model, which was trained using the AdaBoost classifier. The model showed encouraging results, attaining high precision and accuracy of 0.98 and 0.96, respectively. Additionally, the suggested model's resilience is thoroughly evaluated using cross-validation, which shows constant performance throughout all rounds. Dip et al. (Saha et al., 2024) developed an ensemble of hybrid model-based methods, a robust detection model to determine whether or not people are depressed. The outputs of the classifiers are based on both independent and dependent variables that are present in the fresh data set utilized for learning the meta-classifier, following training and testing using the base method. By using the SMOTE in conjunction with cluster sampling, the training data was decreased, increasing the accuracy of depression detection. Additionally, replicating data points can successfully lower the danger of over-fitting. The accuracy of the suggested ensemble technique was the highest, at 99.39%, with an F1 score of 99.51%. Fahmida et al. (Khanom et al., 2024) proposed an explainable ensemble ML approach to preserve the high predicted precision while offering local and global interpretability in the diagnosis of PD. In the research, two clinical datasets were used, which were meticulously selected and refined using a two-step

data preprocessing method that managed outliers and guaranteed data balance, thus minimizing bias. The accuracy was consistently highest when using a stacking model in conjunction with LDA feature optimization and several ensemble ML models. Described AI techniques were also incorporated into the stacking model to ensure transparency. By using these techniques after testing, they made accurate predictions and thorough justification.

To evaluate various features in a multi-agent system, seven classification techniques K-nearest neighbor (KNN), decision tree classifier (DTC), random forest classifier (RFC), MLP, the CNN model, and a hybrid model were used as classification algorithms. This research uses an extremely non-intrusive method that analyzes a set of variables obtained from patient voice analysis. One such kind of the largest drawbacks just using dataset is the difficulty in accurately identifying the entire modeling scale and muscle scale voice information. Because auto-encoders simplify the data by removing potentially noisy variables from the data, we used them in our work and took advantage of the PD-CNN models capacity to obtain more details from the data. Currently, a simultaneous classification and multi-class system has been proposed, that enables the ensemble of PD-CNN model more effectively identify the most pertinent characteristics to make the most accurate prediction possible. In this research, a hybrid model called ParkRLP and MLP represent the two fundamental techniques that comprise the proposed ensemble approach. Both the ParkRLP and MLP methods are learners at level 0, whereas the lightweight PD-CNN model is a level-1 learner. There are eight layers in the compact PD-CNN model. The conv-1D, Maxpooling-1D, and Dense layers have been used to modify the PD-CNN model. According to the dependent and independent factors in the newly generated dataset, the meta-classifiers were utilized for training. The outputs of the methods are determined following their training and testing at level 0. The contributions of this work are as follow:

1. We proposed an ensemble strategy that combines lightweight CNN with ML techniques for the early diagnosis of PD.
2. To create the new dataset, the ParkRLP and MLP networks are level-0 learners, while the lightweight PD-CNN model is a level-1 learner. Successful PD detection is achieved by the proposed ensemble strategy using the level-1 classifier as the PD-CNN model.
3. The lightweight PD-CNN model consists of eight layers, including conv-1D, maxpooling-1D, and dense layers.
4. We incorporate manipulation of data techniques that improve the precision of the ensemble model performance, like SMOTE and cluster sampling.

This paper's remaining sections are structured as follows. Related work is presented in Sect. 2. A more thorough description of the methodology used in the various suggested techniques is provided in Sect. 3. Sect. 4 describes the experiments result, and Sect. 5 presents the analysis of results and comparison. Finally Sect. 6 concludes the paper.

2 Related Works

PD has been identified by numerous researchers by using a various techniques. In light of the current difficulties in PD sub-categorization, risk analysis, and diagnosis using voice signal features, these studies lay a solid foundation for the use of ML and neural networks to detect PD diseases.

Alshammri et al. (2023) used a total of five ML and DL classifiers to recognize PD. The model proposed to use ML and DL techniques to identify PD using speech signal features. The results of these MLP classifiers had the highest accuracy of 98.31%. The dataset, which included 195 recordings of an examination conducted on 31 patients, was obtained from the UCI ML repository. Senjuti Rahman et al. (2023) analyzed and rated the performance of various classification models. Upon applying the different classification model to datasets available in the UCI repository, it was evident that XGBoost performed better than the competitors, attaining an accuracy rate exceeding 92%. Using one DL techniques, a neural network classifier (DNN2) with a reasonable accuracy of 95% was also produced. Sakibur et al. (Sajal et al., 2020) proposed system with voice recorder sensors built into smartphones to collect data on vowel phonation and resting tremors. To create and improve ML models that perform better, data are primarily gathered from patients diagnosed with PD and healthy individuals. When applied independently, voice data and tremor data had the highest PD detection accuracy using offline data, at 98.3% and 98.5%, respectively. KNN provided the highest accuracy over support vectors in both instances. Yaman et al. (2020) suggested using statistical pooling to increase the features of the dataset. Then, using ReliefF, the highest weighted features are chosen from the expanded feature vector. The most heavily weighted feature vector obtained was used for the classification. The SVM and KNN algorithms were applied in the suggested method. SVM and KNN were used to calculate the success rates, which were 91.25% and 91.23%, respectively. The proposed method makes two main contributions to the literature.

Teresa et al. (García-Ordás et al., 2023) suggested a method based on various DL techniques. Finding out if someone has severe or non-severe PD on the one hand, and finding out using regression techniques how far along the disease is in a particular patient on the other. Utilizing both

the motor and total labels of the UPDRS (Unified PD Rating Scale), the mixed MLP, which simultaneously classifies and regresses, produced the best results. The success rate for solving the PD prediction is 99.15%. Sunil et al. (Yadav & Singh, 2021) applied RF, XGB, SVM, and KNN models. Fivefold cross-validation forms the basis of estimated reliability, similar to the ROC curve and confusion matrix do. The higher accuracy of the five algorithms varies between 70 and 95 percent. SVC achieved 93.83% accuracy from the five fundamental classifiers. To identify PD patients early, Wang et al. (W. Wang et al., 2020) used 12 ML algorithms with integrated technology to compare accuracy. In total 183 healthy individuals and 401 early patients were admitted. The classifier has an average maximum accuracy of 96.45%. Two of these methods that are frequently used as ML models are SVM and Artificial Neural Network (ANNs). SVM draws a line that best divides data into classes and offers linear mapping between inputs and outputs. Based on ANNs, recently developed deep-learning techniques, also referred to as deep-neural networks, represent a powerful new suite of tools for PD data classification (Q. Wang et al., 2024; Zhang et al., 2020).

Recently published studies have been incorporated to better compare our results. This shows great promise for improving the accuracy of PD detection. Shawki et al. (Saleh et al., 2024) used two distinct sound datasets along with an ANN and other ML techniques to predict individuals with PD. To obtain the greatest performance and outcomes, they used several strategies, including cross-validation and optimal hyperparameter tuning. The ensemble voting classifier achieved a benchmark accuracy of 97.35%. To identify Parkinson's disease, Sunil et al. (Yadav et al., 2023) used correlation matrices, ML classifiers, and additional tree classifiers as feature extraction techniques. Twenty-three characteristics and 197 incidences make up the PD data set. Of these, eight patients were healthy and 23 are identified as PD patients. In this study, the accuracy of the DT classifier was higher at 94.87%. Using the UCI ML repository dataset, Akila et al. (2024) developed a neural network model to classify PD by interpreting the voice signals. They employed a novel neural network and a unique multi-agent salp swarm (MASS) algorithm to categorize patients with PD. The outcome demonstrated the superior classification accuracy of 99.1% produced by the MASS-PCNN model. Juan et al. (Martínez-Eguiluz et al., 2023) assessed nine methods that used a large set of non-motor clinical PD indicators to separate PD patients from controls. They found that feature selection improved performance and resulted in a large decrease in the number of variables. With an accuracy of 84.4%, their suggested model could identify PD patients by analyzing non-motor factors using ML techniques. Linlin et al. (Yuan et al., 2024) employed ML algorithms using

linguistic signal data as input and integrated the generated classifiers to increase the accuracy of the PD categorization. They used the voice sounds of 252 subjects as a dataset. The results showed that these ML methods might achieve a diagnostic accuracy of up to 95% diagnosis accuracy.

To diagnose PD accurately, Hema et al. (2023) suggested four different classification algorithms using feature selection techniques. The Parkinson's Progressive Markers Initiative (PPMI) dataset was used in the experiments. Numerous assessment techniques are employed to gauge the performance of classifiers. Overall, the RF classifier with the four feature selection techniques produced the best results, with an average accuracy of 96%. For PD prediction using single vowel sound, Wei et al. (Liu et al., 2023) presented a vocal-based ANN model. 48 PD sufferers and 20 healthy people's speech samples were processed by them. For PD prediction, they constructed and assessed an ANN model. With a 91% prediction accuracy, the proposed model performs well. To effectively detect PD, a hybrid CNN-LSTM classifier was presented in this article (Pandey et al., 2024). To evaluate the suggested methodology, 28 Italian-speaking Parkinson sufferers and 22 healthy individuals were used. Using the proposed approach, an average accuracy of 97% was attained. These findings imply that the suggested method suits PD automated identification in real-world situations. To detect and classify PD, Samiappan et al. (Dhanalakshmi et al., 2024) employed several binary classification techniques. They used 192 healthy and 564 PD examples that comprise the PD features dataset. The imbalanced dataset was resolved using SMOTE. A good accuracy of 96.5% was obtained in this study using a combination of SMOTE-ENN and SVM. Sudhansu et al. (Nayak et al., 2023) introduced a genetic approach for feature selection and a novel framework for extracting parameters from audio recordings. This study had a maximum categorization accuracy of 98%. The findings showed that the suggested feature combination can accurately diagnose PD in an individual based only on speech.

In order to gain a thorough understanding that can magnificently detect neural network-based PD diagnoses, more research is therefore required. Better accuracy can be obtained if the complexity of the previous and traditional methods can be reduced. The model and methodology suggested in this study can improve efficiency and robustness as well as pinpoint the scenarios that will accurately identify PD.

3 System architecture and methodology

In this section, we discuss our proposed approach for determining the best outcome from a given dataset. Many ML, CNN, and neural network algorithms are selected for this process. These are the KNN, DT, RF, MLP, CNN, a hybrid model and the proposed ensemble model. We created an ensemble method based on lightweight CNN and ML methods that detected PD with higher accuracy.

3.1 Dataset preprocessing

Data preprocessing is of utmost importance in data analysis and ML. It plays a vital role in ensuring data quality by cleaning, resolving errors, and addressing missing values. An addition, it facilitates feature engineering, which can significantly enhance the performance of ML methods by selecting relevant features, transforming data, and creating new informative attributes. By ensuring that the data are in an optimal state for analysis, data preprocessing paves the way for more accurate insights and robust predictive models.

3.1.1 Null value tracing and label encoding

Checking for null values in a dataset are crucial because they ensures data quality, completeness, and statistical validity, ultimately influencing the performance of ML methods and accuracy of analytical insights. The proper handling of null values is essential for accurate results, and understanding why data are missing can provide valuable domain insights. We traced null values but none were found. Then, the "status" feature was selected as the label and as a binary classification problem, we converted "yes" to 1 and "no" to 0.

3.1.2 Feature scaling

We proceeded to the data transformation phase, wherein we employed the Standard Scaler to standardize each feature independently. StandardScaler is a popular technique for data preprocessing that is used in ML to standardize or scale features in a dataset. It transforms the data such that it has a 0 as the mean and a standard deviation of one. This is performed to ensure that all features have the same scale, which can be important for some ML algorithms, particularly those based on distances and gradients, such as support vector machines and k-means clustering. Certainly, here's the equation for standard scaling is:

$$S_v = \frac{d - \mu}{\sigma} \quad (1)$$

where the initial information is denoted by d , the standardized value by S_v , the mean of the feature values by μ , and the standard deviation of the feature values by σ . A comparison of correlations before and after standard scaling is presented in Fig. 1.

3.1.3 Data split

The train-test divide is a fundamental technique in ML and model validation. This procedure divided a dataset into two separate subsets: a testing set and a training set. A method can acquire different patterns and connections in the information by training. On the other hand, the testing set is kept separate and used to measure the model's efficiency and generalization capabilities. One may evaluate the model's ability to produce accurate projections of new data and unseen cases by assessing its performance using new information from the test set. The train-test split is essential to prevent overfitting, in which a model keeps the training set but fails to extrapolate to fresh data. This contributes to the performance of the model being dependable and appropriate for real-world situations. The division ratio between the training and testing sets can vary, for our dataset we chose 70% of data for training the model and 20% for testing.

3.2 Data manipulation

Data manipulation through oversampling and undersampling are techniques used in ML to address class imbalances in datasets. Oversampling adjusts the distribution by expanding the quantity of occurrences within the minority group, often through synthetic data generation, aiding the model's understanding of the minority class. In contrast, undersampling reduces the majority class instances to avoid model bias but may lead to information loss. The choice between these methods depends on the dataset and algorithm, and

other approaches, such as ensemble methods, should also be used to effectively handle class imbalance. As this dataset has an imbalance, we chose to include data manipulation methods to mitigate the problem and test the performance of the ML and DL models perform. We utilized two methods, SMOTE and cluster sampling in the process.

3.2.1 Applying SMOTE

SMOTE (Al Majzoub et al., 2020) represents the “Synthetic Minority Over-sampling Technique”, which is a common technique for ML to unbalanced datasets (Al-Shaaby et al., 2020). The SMOTE function creates artificial cases of the minority group to equalize the distribution of the classes. This approach involved selecting a statistic about the minority group and figuring out its KNN. Then, by combining across the chosen and adjacent data points, SMOTE generates additional data sources. The method we suggest balances the distribution of classes by using SMOTE to generate synthesized instances of the minority class. It reduces the possibility of over-fitting, which could affect simple data point replication and bias towards the majority class. The results are in a group of artificial sources for information that are similar to the existing minority class instances but introduce some variation. The technique as follows:

$$\mathbf{T}_{\text{new}} = \mathbf{T}_i + (\mathbf{T}_{\text{nn}} - \mathbf{T}_i) \times r_v \quad (2)$$

where the initial minority class instance is denoted by $\mathbf{T}_i$, the new artificial instance is denoted by $\mathbf{T}_{\text{new}}$, a value among the k-nearest neighbors of $\mathbf{T}_i$ is denoted by $\mathbf{T}_{\text{nn}}$, and the interpolation is controlled by r_v , which is a random value between 0 and 1.

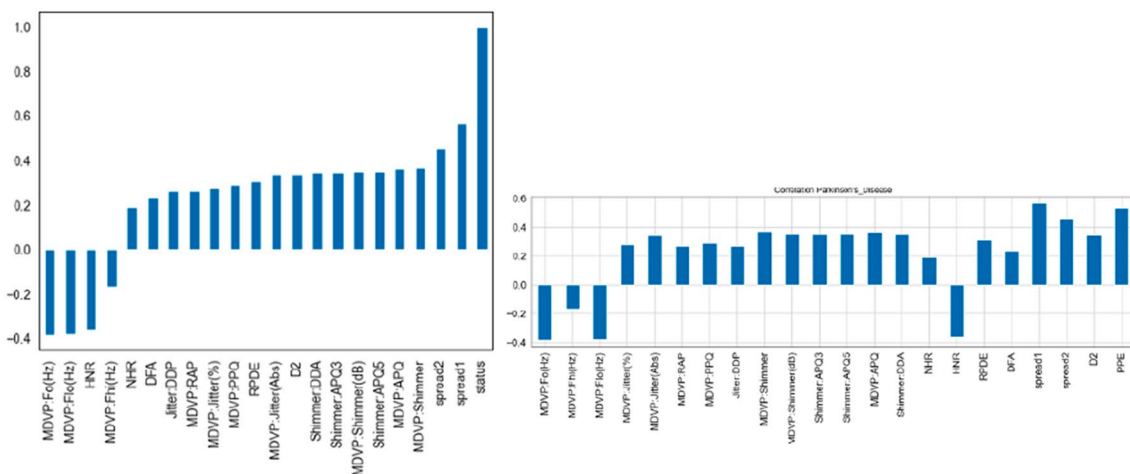


Fig. 1 Correlation before standard scaling and after standard scaling

3.3 Cluster sampling

A statistical sampling approach used in research and survey methodologies is cluster sampling. The population is separated into sections or clusters for cluster sampling, and these groupings are then randomly chosen for inclusion in the collection (Han et al., 2023). Each selected cluster is then further studied in its entirety, which often involves collecting data from all the individuals or units within the chosen clusters. Cluster sampling is particularly useful when it is impractical to directly sample individuals, as it can reduce the cost and logistical challenges of data collection. However, this can introduce variability within clusters, and the choice of clusters should be random to ensure that the sample remains reflective of the target group. In this study, we used it to undersample the imbalanced dataset to test on different ML and CNN models.

3.4 Methods

In this study, to obtain the best results, different types of ML and CNN models were explored. In addition, ANN and ML based hybrid models were used to test whether the proposed model performed better than the single or regular models. And lastly an ensemble model has been proposed in this research.

3.4.1 Artificial neural network (MLP)

MLP is a type of ANN used in ML and DL. This structure consists of an input layer, three hidden layers, an output layer, and several layers of coupled neurons. Every neuron in the neural system processes and passes information to the next layer through weighted connections and an activation function (Tolstikhin et al., 2021). MLPs are ideal for various applications, including regression, and classification, since they can simulate intricate and non-linear relationships in data. Usually, optimization techniques like back-propagation are employed to modify the weights and biases to reduce the variance between the real goals and the anticipated outputs.

The mathematical equation for a MLP method is composed of multiple equations, one for each layer in the network. The following equations represent the single hidden layer forward flow of the MLP.

1. Initial hidden layer input:

$$z_1 = W_1 \cdot X + b_1 \quad (3)$$

Here: - z_1 is the weighted sum of the input features X for the first hidden layer. - The relationships among the source

layer and the initial layer that is concealed are represented by the weight matrix W_1 . - The skew vector of the identifier for the first hidden layer is b_1 .

2. Function of stimulation for the initial hidden layer: The hidden layer is often a non-linear activation function, such as the Rectified Linear Unit (ReLU), and sigmoid.
3. Input for the output layer:

$$z_2 = W_2 \cdot a_1 + b_2 \quad (4)$$

Here: - For the resultant layer, z_2 is the average of the events from the first concealed layer. - The weighted vector for all connections among the final layer and the initial concealed layer is denoted by W_2 . - The resultant layer's skew vector is denoted by b_2 .

4. Output of the MLP:

$$Out_M = \text{softmax}(z_2) \quad (5)$$

The softmax function transforms the original scores z_2 into class probabilities in multiclass classification. The equations above describe a basic feedforward neural network with a single hidden layer. Finding ideal weight matrices (W_1 and W_2) and bias vectors (b_1 and b_2) by modifying them during the training process is the aim of training an MLP. This minimizes a selected loss function, usually associated with the difference between the predicted and true goals (Guo et al., 2021).

3.4.2 Hybrid model (ParkRLP)

ParkRLP is a hybrid model designed to improve PD detection accuracy by fusing the MLP and RFC models. It provides a hybrid approach that makes use of the unique advantages of both traditional ML techniques and NN. In this combination, the MLP handles the duties of feature extraction and complex data representation learning. Simultaneously, the RFC assumes the function of the classifier, drawing judgements based on the features that the MLP has discovered. Because the MLP excels at capturing complex patterns and the RFC is good at successfully separating various classes, this combination is particularly beneficial when working with high-dimensional or non-linear data.

We presented the architecture of the proposed hybrid model ParkRLP in Fig. 2. The approach that uses the output of the final hidden layer as the RFC's input produces a more precise PD prediction. The outcomes are typically better than using the RFC and MLP classifiers independently, but the procedure is a little more complex. The mathematical equation for the ParkRLP technique consists of distinct

Fig. 2 The ParkRLP architecture, hybrid of MLP and RFC. The output of the MLP block connect to RFC block and RFC block predicts PD

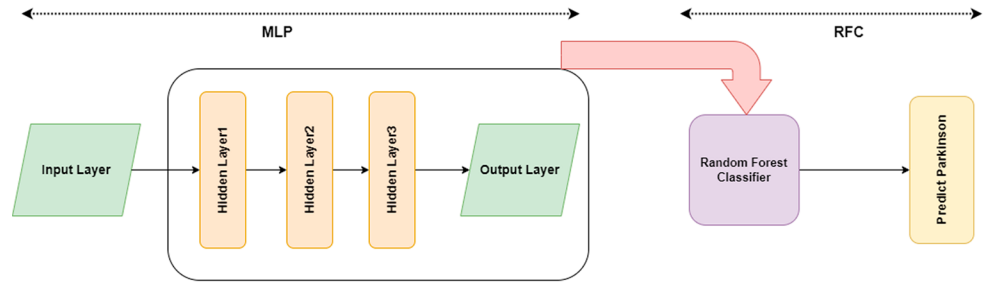
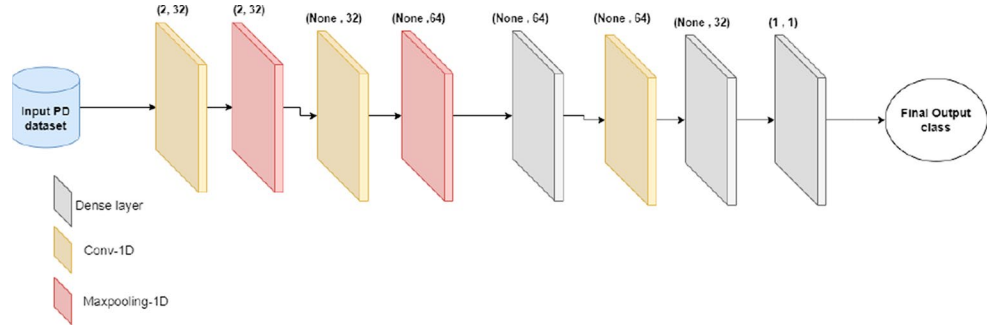


Fig. 3 Modified architecture of lightweight CNN model(PD-CNN) for detection PD. It consists of conv-1D, Maxpooling-1D, and dense layer



formulas for every network layer. The following formulas illustrate an MLP with an input layer and a hidden layer forward pass:

$$Z_1 = W_1 \cdot (a_1 + a_2 + a_3) + b_1 \quad (6)$$

Step 1: The weighted sum of the triggers from the input layer to the hidden layer is denoted by Z_1 for the output layer. The connections across the output layer and the first hidden layer are weighted by W_1 . Additionally, the hidden layers are a_1 , a_2 , and a_3 . The output layer's bias vector is b_1 .

$$Out = \text{softmax}(Z_1) + \frac{1}{N} \sum_{i=1}^N (f_i - y_i)^2 \quad (7)$$

Step 2: The MSE of our data's branching from each node is used when the RFC with the MLP output layer is added. Where N is the number of observations, f_i is the i -th observed value, and y_i is the matching predicted value for f_i . The final output will generate by the equation 7. Results from the hybrid model are more accurate since the final hidden layer of the MLP contains more significant properties for every training set, which is then categorized again using RFC.

3.4.3 CNN model(PD-CNN)

To offset the shortcomings of the traditional CNN model, convolutional and pooling layers are alternated and followed by one or more Dense layers. The convolution layer is balanced by the dense layer that overlays its inputs, which is

composed of many stacked kernels. The input signal obtains high-level properties using a sliding-window method, and the output presents feature maps. By utilizing the pooling operators to aggregate data from every minuscule segment of the input feature systems, a pooling layer provides a traditional downsampling process before selecting the key feature.

In this research, we applied a modified CNN model namely PD-CNN to detect PD at early stages. The light-weight architecture of PD-CNN model presents in Fig. 3. Total of eight layer have been utilized in this architecture. The conv-1D, Maxpooling-1D, and Dense layers have been used to modify the PD-CNN model. The input is presented as $I = [i_1, i_2, \dots, i_C]$, where $x_j \in r^w$ denotes the j -th channel and $j = [1, 2, \dots, C]$. The Max Pooling-1D (M) operation created two distinct channel identifiers, F_M^C and r , where r represents the reduction ratio to accumulate temporal data. Channel-wise statistical vectors are created by forwarding the weighted description to an interconnected network ($ch(V)$), as shown in equation (6) and The channel recalibration vectors ($ch(\bar{V})$) shown in equation(7)

$$Ch(V) = w_2 \times F_M^C(V) + (1 - w_2) \quad (8)$$

$$Ch(\bar{V}) = V \times \sigma \left(f_c^{1 \times C} \left(\delta \left(f_c^{1 \times C/r}(V) \right) \right) \right) \quad (9)$$

A 2×32 convolution procedure using a single channel yields projected values of the total of weighted M . The timewise information matrices were produced using the sigmoid function σ .

Table 1 Description of the modified PD-CNN architecture

Layer	Type	Kernel size	Number of kernels	Output size
1st	Conv-1D	2	32	(2, 32)
2nd	Maxpooling-1D	2	32	(2, 32)
3rd	Conv-1D	None	32	(None, 32)
4th	Maxpooling-1D	None	64	(None, 64)
5th	Dense	None	64	(None, 64)
6th	Conv-1D	None	64	(None, 64)
7th	Dense	None	32	(None, 32)
8th	Dense	1	1	1 (Final output)

$$St(V) = \sigma(f^1 \times 1(w_1 \times (F_r^w(V)) + (1 - w_1))) \quad (10)$$

In Table 1 all the layer and description of PD-CNN model has mentioned. Based on the CNN model, it produces decoupled spatial relations and cross-channel correlations within different CNN module. Without average pooling, data with a single 2×32 convolution size as input result in different 1-D matrix. Before obtaining forward patterns

these convolution sizes operate in distinct portions of the output channels. The Maxpooling-1D connects (none, 64) kernel at the dense layer o first phase. In sixth layer the kernal size was (none, 64) for the same output size. The final layer was 1×1 dense layer to detect PD actual class.

3.5 Proposed ensemble model

The ParkRLP and MLP methods are the level-0 learners in this study, whereas the PD-CNN method is the level-1 learner. The MLP and hybrid methods were selected as the foundational learners (Mienye & Sun, 2023) primarily because of their robustness in such circumstances and outstanding performance in sequence data representation (Saleh et al., 2024). Furthermore, the ensemble method must be diverse since different base models produce alternative mistakes (Shastri, 2023). The proposed procedure is presented in Fig. 4. Algorithm 1 also describes the proposed ensemble method.

Fig. 4 Two basic models work together to generate a new dataset in the proposed ensemble of the PD-CNN method, and the PD-CNN technique acts as a meta-model to provide the final prediction

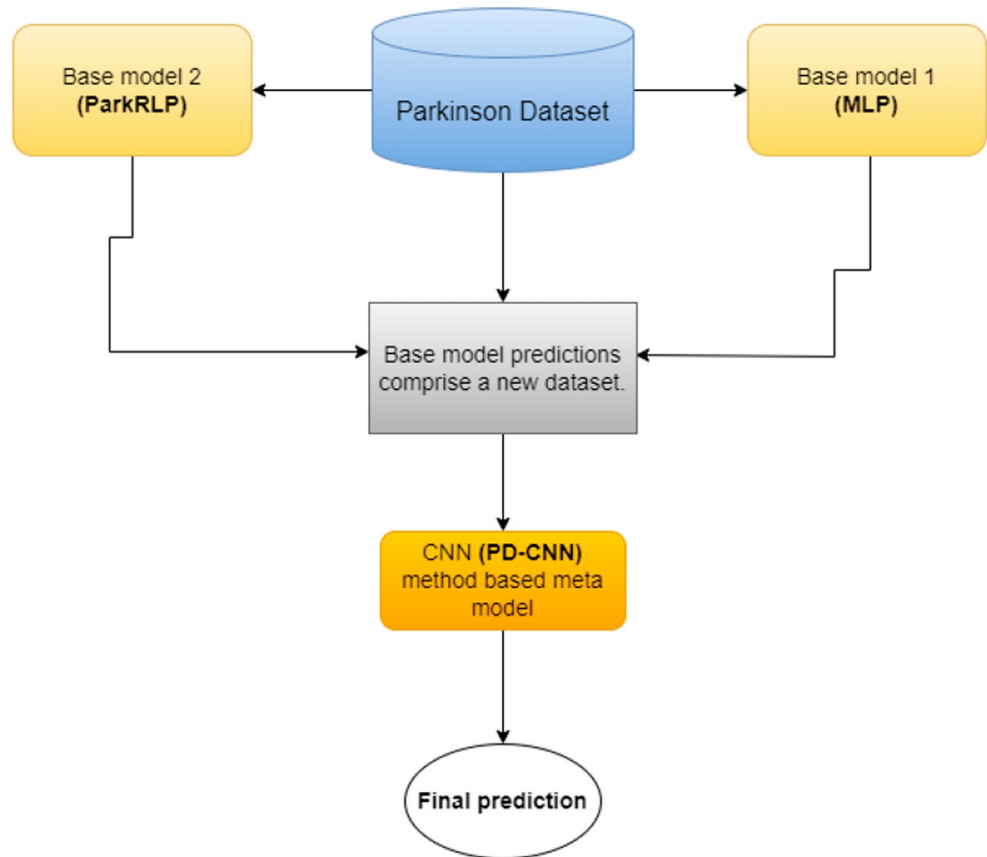


Table 2 Computational resources to implement the proposed model

Environment	Parameters
Platform	Google Colab/Jupyter notebook
Language version	Python 3.0
Graphics card	TPU, 13.7 GB
Available RAM	5.8 GB
Disk space	7.9 GB

Algorithm 1 Proposed ensemble of PD-CNN method.**Input:** PD Dataset, D_t $(c_1, d_1), (c_2, d_2), \dots, (c_j, d_j)$

The selected two base model ParkRLP and MLP

The Ensemble meta learner PD-CNN model

Methodology

Step1: Training the models at level-0.

While $t = 1, \dots, T$ Utilising D_t , train a base classifier m_t Step2: Comprising a new dataset D_N **While** $i = 1, \dots, j$ Create a new sample $\hat{c}_i, \hat{d}_i$ using Equation 11Step3: Adapt the meta-model $\hat{m}$ that is CNN-based using D_N **return** $M(c) = \hat{P}(m_1(c), m_2(c), \dots, m_T(c))$ **Output:** CNN based ensemble meta classifier M

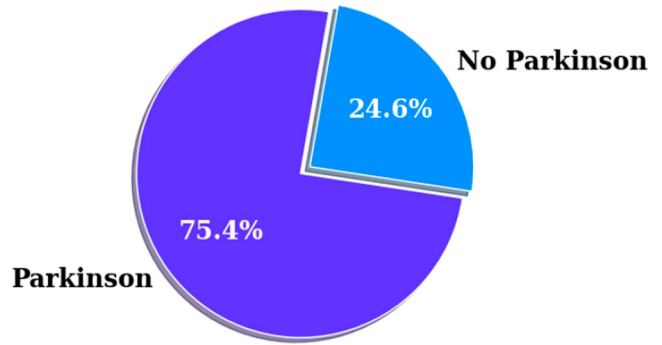
Initially, the base models were trained using the MLP and ParkRLP methods. The ensemble method was created utilizing the two-fold cross-validation (CV) to overcome dataset leaking. By combining the actual labels with the out-of-fold predictions. Then, the second stage generates a new dataset. In particular, the initial category descriptors comprise the outcome parameter in the new dataset. Furthermore, the basic model utilized identical CV indexes over the training set, whereas the CNN technique was trained using the out-of-fold predictions. Since the level-0 methods were not trained on events, the actual labels were attached to them. Therefore, the base methods outputs of 1 and 0 reflect PD-affected and non-PD-affected.

For example, a new sample $\hat{c}_i, \hat{d}_i$ is made if each sample in a Parkinson dataset D_t is c_i, d_i .

$$\hat{c}_i = m_1(c_i), m_2(c_i), \dots, m_T(c_i) \quad (11)$$

where the base models are $\{m_1, m_2, \dots, m_T\}$ and $\hat{c}_i$ is the sample prediction. Finally, in the third phase, the CNN-based approach combined with the base methods to train on the final dataset. The final ensemble prediction prediction for c is $\hat{m} \{m_1(c), m_2(c), \dots, m_T(c)\}$, where $\hat{m}$ represents meta model.

Does the person has parkinson's Disease

**Fig. 5** Ratio of the dataset before applying SMOTE

4 Experimental evaluation and result analysis

In this work, we experimented with the models in four different scenarios and tested the performance of the models from these. The scenarios are preprocessed data applying only SMOTE, and preprocessed data with SMOTE and with Cluster Sampling combined.

4.1 Specifications of the environment setup

Graphics Computing Units (GPUs) (Corral et al., 2024) or central processing units (CPUs) can provide a significant amount of the processing power required for machine learning classification. A GPU or CPU is required to support the processing activities. As a result, we trained all of our models using Google Colab (Kuroki, 2021), which provides access to powerful cloud GPUs. Table 2 lists the parameters and configuration of the environment we used for our inquiry. A v3 TPU processor is included with the Google Colab. To implement our suggested model on the Google Colab or Jupyter Notebook platform, we require 5.8 GB of RAM, 7.9 GB of disk space, and two TensorCores. The ensemble of PD-CNN models in a computer system can be trained using these criteria.

4.2 Dataset

The PD dataset (Alshammri et al., 2023) used in this research. It contains various biological voice measurements of 31 persons, out of them, 23 persons carry PD (Little, 2008). The ratio of the dataset mention in Fig. 5 Each of the rows in the dataset corresponds to one of the 195 voice recordings from these individuals, and each column represents a specific voice measurement, identified by the column “name”. This dataset is represented by the “status” column, which has a value of 1 for PD and 0 for healthy.

Table 3 The features and definitions of the Parkinson dataset

Measure of speech	Definition
Name	Subject's name in ASCII and the audio number
MDVP:Fo(Hz)	The typical voice frequency response (numerical parameter)
MDVP:Fhi(Hz)	Highest frequency response of the voice (numerical factors)
MDVP:Flo(Hz)	Least frequency response of the voice (numerical factors)
MDVP:Jitter(%), MDVP:Jitter(Abs), MDVP:RAP, MDVP: PPQ, Jitter:DDP	Multiple frequency variation metrics (numerical factors).
MDVP:Shimmer, MDVP:Shimmer(dB), Shimmer: APQ3, Shimmer: APQ5, MDVP: APQ, Shimmer:DDA	A number of magnitude variance metrics (numerical factors).
Status	0 for HC and 1 for PD
NHR, HNR	measurements of the voice's sound tonality ratio (numerical factors).
Status	Status of health condition (1) - PD, (0) - healthy
RPDE, D2	Measures of dynamic behavior complexity (numerical factors)
DFA	Coefficient of frequency fractal scaling (numerical factors).
Spread1, Spread2, PPE	Dynamical metrics of basic frequency range (numerical factors).

Table 3 provided a description of the key characteristics of the features. Overall, this dataset is a decent one that holds actual data of PD and normal patients and can be used to classify between those.¹

4.3 Model evaluation metrics

Accuracy A_c in ML is calculated by dividing the number of accurate predictions made from a model by the entire number of predictions, providing a straightforward measure of its overall correctness (Mulla & Gharpure, 2023).

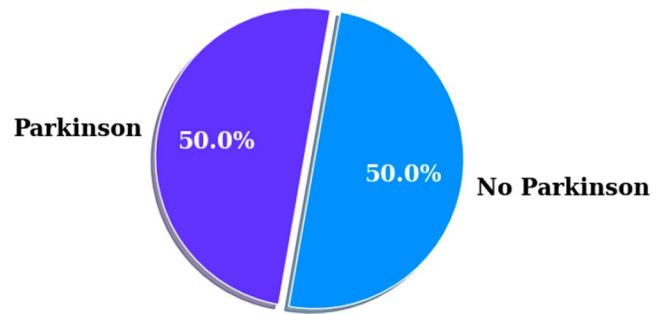
$$A_c = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

Precision The P_r statistic assessed the ratio of true positive forecasts to all optimistic predictions. This parameter shows how well the classifier can identify positives while reducing the amount of false positives.

$$P_r = \frac{TP}{TP + FP} \quad (13)$$

¹ <https://archive.ics.uci.edu/dataset/174/parkinsons>

Does the person has parkinson's Disease

**Fig. 6** Data ratio after applying SMOTE**Table 4** Model score on dataset after applying only SMOTE

Models	Accuracy	F1-Score	Precision	Recall	Log Loss
KNN	92.59%	93.67%	93.50%	93.40%	7.12
Hybrid model (PakRLP)	96.59%	96.74%	97.00%	97.00%	3.18
MLP	95.89%	96.00%	94.40%	95.20%	6.36
DTC	93.04%	93.25%	90.91%	95.31%	6.73
RFC	93.92%	93.92%	94.05%	93.63%	6.51
PD-CNN	97.41%	98.00%	98.00%	97.67%	2.95
Ensemble of PD-CNN	98.12%	99.00%	98.57%	98.19%	2.43

Recall: The R_c parameter indicates the proportion of true positive outcomes in the dataset that were positive forecasts.

$$R_c = \frac{TP}{TP + FN} \quad (14)$$

F1-Score: The F_s combines both R_c and P_r into a single number, so achieving a compromise. It is especially useful when trying to determine a trade-off between producing accurate positive forecasts and documenting every positive case (Kumar Saha et al., 2024).

$$F_s = \frac{2 \cdot P_r \cdot R_c}{P_r + R_c} \quad (15)$$

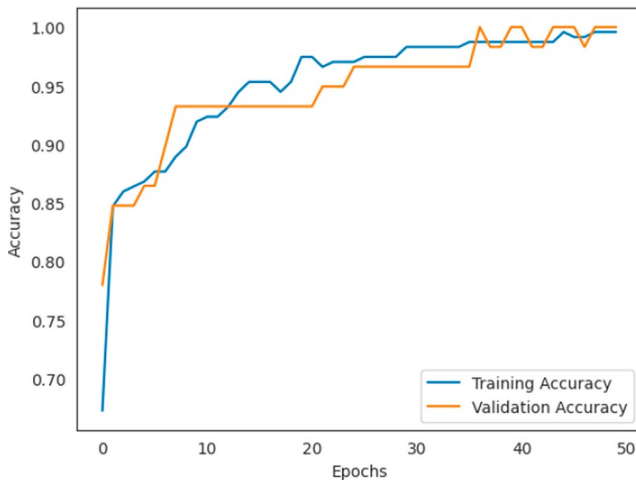
4.4 Results applying SMOTE

The issue with this dataset is, that it has an imbalance in it. 75.4% of data shows healthy and 24.6% data shows PD. To balance it, an oversampling technique can be applied using SMOTE. After preprocessing, SMOTE is applied and this method generates synthetic data to balance the dataset which can be seen in Fig. 6.

After balancing the dataset we passed it for training and the results are shown in Table 4. The Table demonstrates a notable enhancement in results following the application

Table 5 Model score on dataset after applying SMOTE and Cluster Sampling combined

Models	Accuracy	F1-Score	Precision	Recall	Log Loss
KNN	94.92%	94.92%	96.55%	93.33%	5.08
Hybrid model (PakRLP)	98.31%	98.81%	98.45%	99.00%	0.23
MLP	97.45%	97.00%	97.53%	97.50%	1.69
DTC	96.61%	96.77%	93.75%	96.50%	3.39
RFC	93.22%	93.33%	93.33%	93.33%	6.78
PD-CNN	98.45%	98.76%	99.00%	99.00%	1.69
Ensemble of PD-CNN (Proposed)	99.47%	99.50%	99.62%	99.50%	0.89

**Fig. 7** Training curve of proposed PD-CNN model

of SMOTE. When considering the balanced dataset, both the hybrid model and the proposed ensemble of PD-CNN exhibit exceptional accuracy of 98.12%, marking the highest accuracy ever achieved on this UCI dataset. The PD-CNN model displays encouraging performance on balanced datasets.

4.5 Results applying SMOTE and cluster sampling together

After testing individually for Cluster Sampling and SMOTE, we combined both models to test the performance. First, we used SMOTE to oversample the dataset, and then used cluster sampling and SMOTE together. The results for this dataset are presented in Table 5. After this testing, we can see from the table that, out of the seven models, four models performed the best. Here, also the ensemble of PD-CNN model achieved the highest accuracy of 99.47%. This signifies the combination of oversampling and undersampling technique can manipulate the dataset to a level where models such as Pd-CNN and hybrid model performs exceptionally

Table 7 Comparison table with our works and others

References	Applying SMOTE	Dataset	Best Method	Accuracy
Hema et al. (Hema et al., 2023)	No	PPMI	RFC	96%
Samiappan et al. (Dhanalakshmi et al., 2024)	Yes	Private	SMOTE-ENN and SVM	96.5%
Sunil et al. (Yadav et al., 2023)	No	UCI	DT	94.87%
Rasheed et al. (Rasheed et al., 2020)	No	UCI	BPVAM, BPVAM-PCA	97.50%
Akila et al. (Akila & Nayahi, 2024)	No	UCI	MASS-PCNN	99.1%
Raya Alshammri et al. (Alshammri et al., 2023)	Yes	UCI	MLP	98.31%
Senjuti Rahman et al. (Rahman et al., 2023)	No	UCI	DNN2	95%
Jain et al. (Jain et al., 2021)	Yes	UCI	DNN	91.5%
Our Proposed Model	Yes	UCI	Ensemble of PD-CNN	99.47%

well. The model performed remarkably well once the dataset was balanced after applying SMOTE, which had previously caused the dataset to be unbalanced.

4.6 Results applying custom CNN model

This dataset was also examined using a unique tiered model. The model has four layers. An input layer, a dense layer with ReLU activation and 64 neurons, a dense layer with 32 neurons with ReLU activation, and one output layer with Sigmoid activation in the end. Following model construction, we employed the binary loss of cross-entropy function, optimizer Adam, and batch size of 32, and 50 epochs. Then we trained the model on preprocessed non-manipulated data. The training curve is given at Fig. 7.

5 Result analysis and findings

We experimented with various models on this dataset to determine which model performs the best to detect PD from voice signal data. PD is a disease that is treatable if it can be identified at an early stage. We have tested on eight different models. These models are tested on two different conditions,

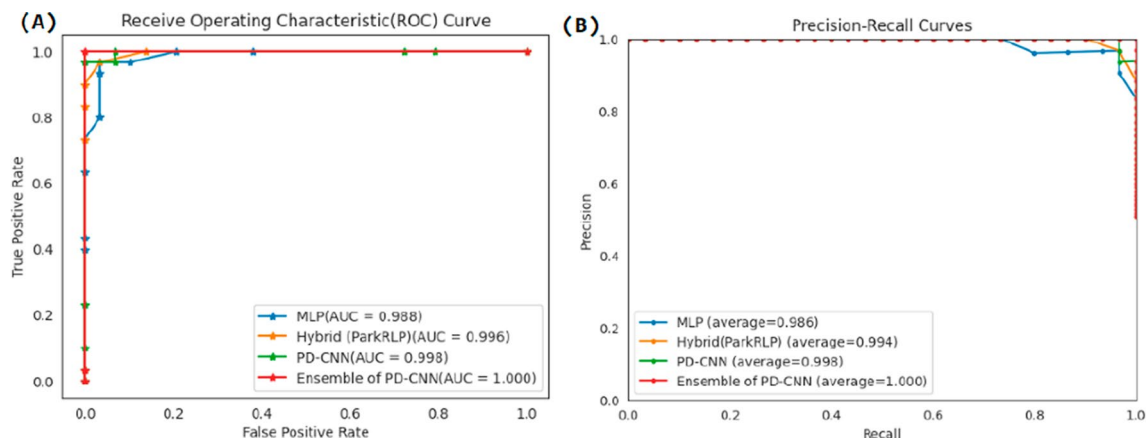


Fig. 8 (A) ROC curve, and (B) Precision-Recall curve for all best evaluated **Methods that combine cluster sampling and SMOTE**

which are dataset applying only SMOTE and combined of SMOTE and Cluster Sampling.

5.1 Classifiers performance matrix

Table 6 showed the AUC score of all models, which is a policy of measures the accuracy of models. The proposed ensemble of PD-CNN model achieves the highest auc score of 100.00%, with CNN models following closely at 99.00%. We can see our proposed ensemble model consistently delivers top-notch results across all conditions, contributing to its highest average accuracy. From these observations, it's clear that proposed ensemble of PD-CNN model hold a promising performance in PD detection.

The confusion matrix C_m for every classifier contains these data points. Figure 8 displays the for the all best evaluated model recommended and used in this architecture. It displays the C_m that the ensemble of PD-CNN model that was suggested produced for the classification. The C_m demonstrates ensemble model achieved nearly 100% accuracy which is a significant contribution of this research.

5.2 Results of classifier parameter

We noted in the previous section that the CNN model ensemble fared better than the other method in this investigation. The outcomes exceeded those of earlier research when we combined SMOTE and cluster sampling. According to Fig. 9, the proposed ensemble of CNN and PD-CNN method detects PD with the highest accuracy. A visual method of evaluating a binary classification model's efficiency is the Receiver Operating Characteristic (ROC) curve. At various limits for the model's projected probabilities, it shows the trade-off between the true positive rate and the false positive rate. The ROC and precision-recall curves of the model types are given at Fig. 10. It demonstrates that, according to our evaluation, ensemble of PD-CNN is the

best-performing model. This ensemble classifier performs better predicting PD than the other models we used for this investigation. Furthermore, we observe that the AUC score of the proposed model is greater than other classifiers. As a result, it is the best model to forecast PD more accurately.

In Table 7, we present a comprehensive comparison of recent studies on this dataset, assessing their respective output. Our proposed methodology consistently outperformed the other models, showcasing the highest accuracy across the board. Particularly noteworthy is the success of our proposed ensemble of PD-CNN model, achieving a higher accuracy of 99.47%. This holds significant promise for early-stage PD detection through analysis of voice signals, potentially playing a crucial role in saving lives by enabling timely intervention for this condition.

6 Discussion

ML technology is the most advanced technological innovation for detective modeling in the area of health research. Thus, the main goal of this research has been to create an automated system that uses speech characteristics to identify if a person has Parkinson's disease. To detect people who might suffer from PD, this paper compares and contrasts several ML and CNN classifiers as well as a hybrid model called PakRLP. With detective accuracy, precision, recall, and f1-scores of 99.47%, 99.62%, 99.50%, and 99.50%, respectively, the model that we have provided an ensemble of PD-CNN performs exceptionally well when considering a variety of performance metrics. The proposed ensemble model helps to mitigate overfitting when we comprise a new dataset using the two base models (ParkRLP and MLP). The final prediction results of the proposed ensemble model rely on the new dataset, and the model performs remarkably well to avoid overfitting issues. Medical practitioners can use the patients' speech characteristics to understand the model's

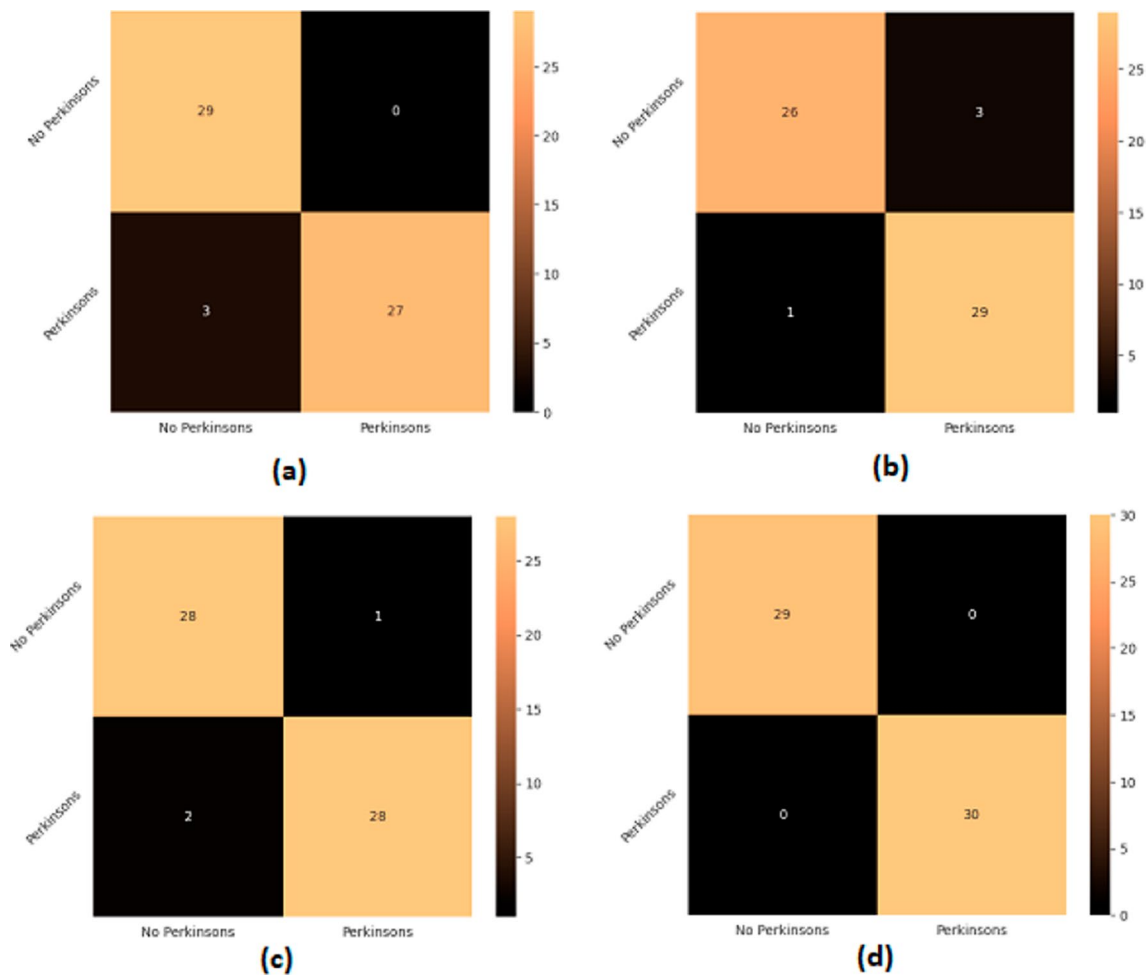


Fig. 9 C_m of all the best performing models combining cluster sampling with SMOTE have shown, (a) MLP, (b) DTC, (c) Hybrid method (ParkRLP), (d) ensemble of PD-CNN model

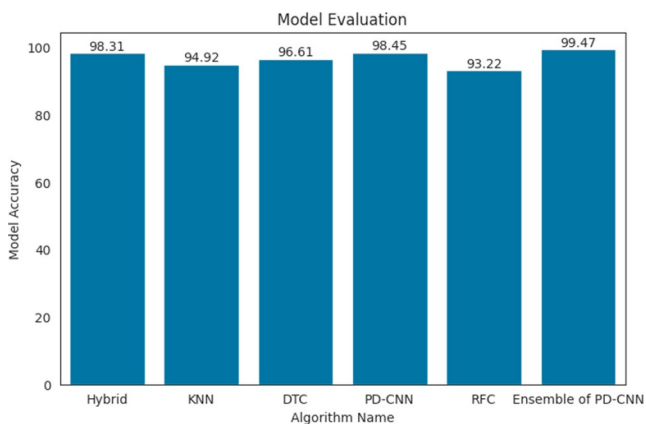


Fig. 10 Evaluation of best model combining cluster sampling with SMOTE

prediction, and then apply these speech signals on the base two models to identify Parkinson's disease. We can conclude by claiming that this will help researchers and medical

Table 7 The AUC score combining cluster sampling with SMOTE of all evaluated methods

Model Name	AUC score %
KNN	98.00
Hybrid model (ParkRLP)	99.60
MLP	98.80
DTC	99.00
RFC	95.00
PD-CNN	99.80
Ensemble of PD-CNN (Proposed)	100.00

professionals diagnose PD issues in real-world clinical trials faster.

7 Conclusion and future work

In summary, the successful identification of PD using patients' voice measures represents a significant advancement in medical diagnostics. MLP and a hybrid model called ParkRLP are the fundamental techniques that combine the

proposed ensemble approach. While the lightweight PD-CNN model is a level-1 learner, the ParkRLP and MLP networks are level-0 learners. There are eight layers in the compact PD-CNN model. Conv-1D, maxpooling-1D, and Dense layers were applied to create the PD-CNN model. According to the dependent and independent factors in the newly generated dataset, the meta-classifiers were utilized for training. After that, the classifiers are trained and assessed at the base level, and their outputs are determined. To enhance novelty, we applied SMOTE only and SMOTE with clustering sampling to the dataset. Using the level-1 classifier as the PD-CNN model we proposed an ensemble approach to detect PD successfully. Within this framework, an ensemble of PD-CNN models has been proposed, which achieved 99.47% accuracy by utilizing cluster sampling and SMOTE together. The proposed methods efficacy was also shown by its consistent 5% advancement in earlier studies on individual CNN and ML models. This model has the potential to support early diagnosis and intervention by medical professionals, ultimately enhancing the standard of living for PD patients.

This study examines and contrasts various ML and ANN algorithms to identify individuals with Parkinson's disease (PD) in their early stages. One of the initial drawbacks is that this study only used the UCI dataset for detecting PD. Furthermore, the number of datasets used in this experiment was inadequate, and clinical datasets involving PD patients were not taken into account with the help of medical specialists. For the improvements, we aim to develop a real dataset with the help of medical professionals for early-stage PD detection. The proposed model will perform better in the real world. When we create the ensemble technique utilizing the MLP and hybrid approach, the lightweight CNN serves as the meta-model to create a new dataset. The lightweight architecture of our proposed ensemble approach will improve performance, even if our goal is to create an actual dataset with differences in patient demographics, recording settings, and voice quality. Therefore, the main goal of this research has been to create an automated system that can assess a person's speech characteristics to detect whether or not they have PD. We intend to use various dimensionality reduction algorithms and consider several clinical parameters while pre-processing a sizable dataset in the future. Additionally, we aimed to develop a real-time expert system to assist physicians in more accurately identifying PD at an early stage.

Author contributions D.K.S Writing – original draft, Writing – review & editing, Methodology, Validation, Software, Investigation, Formal analysis, Conceptualization. T.D Writing – original draft, Software, Investigation, Formal analysis.

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Declarations

Competing interests The authors declare no competing interests.

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